

Program overview

14-Dec-2022 11:21

Year 2022/2023
Organization Industrial Design Engineering
Education Minors Industrial Design Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Minors IO 2022							
IO-MI-075	IO-MI-075-22 Study Abroad						
IO-MI-076	IO-MI-076-22 Workplacement						
IO-MI-124	IO-MI-124-22 Interactive Environment						
IO-MI-221	IO-MI-221-22 Advanced Prototyping						
IO-MI-222	IO-MI-222-22 Designing Sustainability Transitions						
IO3831-19	Demystify Green	3					
IO-MI-223	IO-MI-223-22 Connected Creativity						

Year	2022/2023
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

Minors IO 2022

Year	2022/2023
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-075

Year	2022/2023
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-076

Year	2022/2023
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-124	
Contact for students	Aadjan van der Helm - a.j.c.vanderhelm@tudelft.nl
Program Title	Interactive Environments Minor
Program Coordinator	Aadjan van der Helm - a.j.c.vanderhelm@tudelft.nl
In association with the Faculty of	BK EWI IO
EC Program	30 ECTS
Contact for Students	Aadjan van der Helm - a.j.c.vanderhelm@tudelft.nl
Exit Qualifications	The student will learn to: - formulate an ambitious yet feasible spatial design vision, project planning and strategy, accordingly plan a systematic approach to the design project characterized by the use of theories, models and coherent interpretations, while having a critical attitude, and insight into the nature of most recent science and technology - recognize and apply specific technical knowledge and skills required to complete the main assignment - integrate and employ individual, major-specific skills and knowledge in a product-oriented group work environment - develop and apply task-oriented group working skills and organization strategies, prove competence in cooperating and communicating; be able to not only adequately interact, but also have a sense of responsibility, and leadership - develop and apply skills in using images and written and spoken word in order to convey the essence of a design to others - practical skills for real-life project development - estimate impact of proposed design on culture, society and environment - deploy and user-test preliminary prototypes and the final design product - gather knowledge and insights from developing working prototypes, being both building components and industrial products - present and defend design concepts - understand the concepts of large-scale embedded networks, spatial computing and distributed interaction - evaluate minor activities and compose an essay answering a question and formulating answers to that question
Program Structure 1	The program structure consists of 5 segments. The first segment is about becoming familiar with interaction design. A field trip is organized to Ars Electronica Festival (in Linz Austria), one of Europe's significant festivals on interactive media art. In addition further research is performed into relevant topics that is presented to all minor students. The topic of the second segment is about the design and prototyping of an interactive object. The third segment is about a network of interactive objects. In the fourth segment students design and build a full scale interactive environment, Lastly the fifth segment involves fine-tuning of the interactive behaviour of the interactive environment. At the end of the minor students organize a one week exhibition presenting their final concepts to the public.
Administration by the Faculty of	IO
Minor Title	Interactive Environments
Intended for	The Minor will be offered to TU Delft Bachelor students from all faculties.
Expected prior Knowledge	Students are recommended to have interest in digital technologies as well as capability of creative thinking and solving out of the box problems.
Prerequisites Minor	Students are expected to represent a level of knowledge and skills adequately corresponding to the field of their major. All minor courses are taught in English, fluent command of spoken and written English language is required.
Minor Coherence / Goal	The main field of study in Interactive Environments Minor is multi-user interaction with smart spatial environments. The course structure is built around a group project assignment of an interactive social space, consisting of autonomously operating intelligent building components and smart products. The group project provides a framework which encompasses all course modules and individual student assignments. This delivers a balanced setup between education and practice, geared towards helping students in acquiring practical skills and knowledge needed for building, testing and evaluating experimental interactive complex spatial environments.
Minor Content 1	The minor consists of 5 courses that aim to build skills and involve design activities combined with making prototypes. In the main design course students are encouraged to formulate challenging design visions. Those visions are subsequently realized and tested in form of robust, fully operational and full-scale prototypes. Students from different faculties work together in multidisciplinary teams that design, produce, and deploy unprecedented interactive spatial installations. They receive professional guidance from specialists from involved fields and direct, hands-on access to cutting-edge technologies provided by affiliated commercial partners. In process of building the prototypes they acquire practical knowledge needed to undertake this challenge. This knowledge ranges from creative design strategies through project management skills to technical and technological means of physically constructing a complex interactive structure.
Minor content 2	This is a collaboration course. In this course you work with an external partner. This partner will get the rights to the intellectual property of the results of this course. You can find more information via the IDE student portal (https://www.tudelft.nl/en/student/faculties/ide-student-portal/), under Intellectual Property in IDE education.
Education Methods	The minor involves a mix of group work and individual work. The group work involves (roughly 26 ECTS) collaborating in research, design, programming, constructing and testing. The group work is performed in various team compositions from small teams (2 students) to large teams (10 students). The individual work (roughly 4 ECTS) involves acquiring technical skills and writing essays.
Maximum number of participants	40
Remarks Maximum number of participants	Selection takes place by means of a random draw, the following formula is used: 1/3 students of the TU Delft Faculty of Industrial Design, 1/3 other TU Delft disciplines from Constructive or Design Engineering related programs, 1/3 other disciplines. This allocation is done to meet the interdisciplinary nature of the minor . Minors of the IDE faculty are open only for students from academic programs.
Minimum number of participants	40

Year	2022/2023
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-221	
EC Program	30
Program Start (Study Year)	2019
Program Goals	The Advanced Prototyping Minor brings together the most recent digital manufacturing methods and emerging materials in creating functional high-fidelity prototypes. Prototypes are used in various design and engineering disciplines as an early version of a product or a system. They are built for a variety of purposes including to test an idea or concept (e.g., proof-of-concept prototypes), to represent certain aspects of an intended design (e.g., visual/ functional prototypes), and to trigger discussion in a multi-disciplinary design team (e.g., boundary objects). The course aims at equipping you with a wide range of advanced prototyping skills, techniques and materials to create prototypes, which capture both function and appearance of the intended design. To that aim, the course covers a broad interdisciplinary foundation, in-depth knowledge and practical skills, through (guest) lectures, hands-on workshops and multidisciplinary research/design projects. In addition to the knowledge and skills necessary for building prototypes, the course injects necessary critical and creative thinking to determine which of these techniques and when in the design process could be applied, by considering economics, feasibility, fidelity as well as environmental sensibility.
Intended for	Advanced Prototyping Minor suits both design-oriented students with science/technology background and science/technology oriented students with art/design background. The aim is to build project groups by combining students from these different backgrounds to create an environment in which students can reflect on the contribution of each discipline for advanced prototyping. Selection takes place by means of a random draw, based on the following formula: 1/2 from Industrial Design Engineering (TU Delft) 1/2 from other (Engineering) disciplines (TU Delft) This allocation is done to meet the interdisciplinary nature of the minor. Minors of the faculty of Industrial Design Engineering are open only for students from academic programmes.
Minor Content 1	What's in it for you? - Lectures and guided workshops on a variety of digital fabrication methods and materials processing - Collaboration with other disciplines, including engineering, science, arts, and design. - Access to and use of the facilities/labs, including the Digital Fabrication Lab (DiFaLab), Emerging Materials Lab, the Body Lab (digitalisation @IDE), and Fablab at the TU Delft Science Center.
Minor content 2	Course content: Q1 - Inspirational lectures: The minor starts with introductory lectures on prototyping theory, basic skill training and inspirational (guest) lectures on prototyping in practice. - Knowledge/skill building courses: The two courses Prototyping with/for Digital Fabrication (PDF), and Prototyping with/for Emerging Materials (PEM) will run in parallel in Q1. These emphasize the role of these technologies and materials in prototyping advanced responsive, adaptive and/or ultra-personalized prototypes (i.e., prototyping with), and the role of prototyping for showing the potential of these material and fabrication technologies (i.e., prototyping for). - Workshops: To apply the learned skills and to reach concrete and assessable results, lectures are accompanied by hands-on workshops. - In PDF course, you will be exposed to methods and techniques on digitalization, design automation and digital fabrication, including 3D scanning technologies, 3D modeling and parametric design tools (Rhino Grasshopper), (multi-material) 3D printing, laser cutting, and CNC milling. - In PEM course, you will be exposed to the science and art of working with emerging materials, with a different focus every year, which can be smart materials (e.g., Shape Memory Alloys, smart textiles), or living materials (e.g., Algae) in designing responsive behaviours with these materials. Q2 - Advanced Prototyping project: The obtained knowledge and skills in Q1 are synthesised in the Advanced Prototyping Project. An iterative design process is followed, in which you will design and build (multiple) prototype(s). In addition to constructing prototypes, you will be testing and evaluating the prototypes and eventually exhibiting both the process and the prototypes at a public venue. - in Q2, you will also follow several transferable skills workshops, such as (prototype) photography, video making and pitching. Moreover, there will be several workshops on material and user testing of prototypes.
Minor Assessment	All three courses in the minor are assessed through either individual or group assignments. These will be graded using a rubric. The basic skills training will be graded with a pass/fail. The grading of the assignments will be conducted by lecturers/coaches of the course. Benchmarking will take place in the case of multiple, parallel coaches and between courses to ensure consistency. A student will have to have passed at least 4/6 sub-modules of Q1 to be able to start the project in Q2. The basic skills training will have to be completed before starting the project in Q2. Every individual course will have to be passed with at least 6/10. The final grade will be a weighted average of the course graded (based on EC).
Maximum number of participants	40

Year	2022/2023
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-222	
Minor Coordinator	Ir. C.P.J.M. Kroon
Contact for students	Dr.ir. Jotte de Koning Jotte.dekoning@tudelft.nl
Program Title	Designing Sustainability Transitions
In association with the Faculty of	IO
EC Program	30
Program Start (Study Year)	2019
Program Goals	The minor is characterised by a dual approach to sustainability transitions. The courses stimulate high-level critical thinking through lectures, talks by external experts and group discussions. On the other hand stimulating the development of practical design skills for sustainability through design courses, design masterclasses, and working in multi-disciplinary student teams on the integral design challenge. The learning goals are: - Analyse the different elements of complex sustainable systems, in the context of the challenge of addressing societal needs within our planetary boundaries. - Evaluate processes of change towards a sustainable future at different scales, from individual users to complete societal systems. - Form arguments for your own perspective on how transitions could take place. - Work with at least three different design approaches for sustainability. - Select from a wide range of state-of-the-art design tools and techniques, introduced by experts in guest lectures and workshops. - Apply different design approaches, tools and techniques in practical projects. - Gain experience in carrying out project work in multidisciplinary teams. - Apply communication, entrepreneurship, and collaboration skills towards sustainability challenges in the real world.
Program Structure 1	Throughout the semester you will work in teams composed of students from diverse educational background. A Design Project is combined with lectures, expert talks, workshops, group discussions and field trips. There is cross fertilisation of knowledge and experiences between the courses, they cannot be taken separately. This minor consists of 6 courses: - The Design Challenge (12 ECTS, Q1&2) - Sustainability Issues and Societal Change (3ECTS, Q1) - Value Sensitive Design (3ECTS, Q1) - Design Approaches (5ECTS, Q1) - Masterclasses (4ECTS, Q2) - Demystify Green (3ECTS, Q2)
Administration by the Faculty of	IO
Intended for	Students from Industrial Design Engineering (IDE) and other TU Delft faculties, as well as from design and engineering backgrounds, on a level comparable to TU Delft.
Minor Content 1	What would a sustainable future look like? And how can we design for it? Developing your own answers to these two questions will be at the centre of this minor. Political leaders (agreed, not all), business leaders and members of society recognize the sustainability crisis that we are in; climate change, loss of biodiversity, the threat it poses to societies, economies and our individual well-being. This minor will help you understand the way of thinking that is needed to establish change and the part you can play in responding to the challenges in front of us and ahead of us. You will learn how to use a design approach when dealing with sustainability transitions. You will learn to use specific design approaches and methods to contribute to, facilitate or lead (?) sustainability transitions. You will be stimulated to develop your own vision on sustainability (one of the so-called wicked problems) and toolkit for sustainability that fits with your own views and values. This minor will help you improve your career opportunities: international organisations see the move towards a sustainable world as a complex, multidisciplinary challenge. The minor also collaborates on exploratory projects. You will gain unique experience working on leading topics with some frontrunner organisations and larger organisations. You will learn about different concepts that are central to designing sustainability transitions. Such as transition management, complex systems, systemic design, planetary boundaries, futuring, social design, everyday (social) practices, critical materials, worldviews, values and ethics, behaviour change, participatory design and co-creation and the circular economy.
Maximum number of participants	60
Remarks Maximum number of participants	Minors of the IDE faculty are open only for students from academic programs.

IO3831-19	Demystify Green	3
Course Coordinator	Dr.ir. S.F.J. Flipsen	
Course Contents	Companies and organisations are well aware of the importance of sustainability. They often advertise their contributions to lowering the environmental burden of their products and services. These contributions are not always substantiated and - sometimes- they are simply dishonest (we call this greenwash). In this course you are going to investigate a sustainable claim found in current media. This could be a claim found on Facebook, Twitter, newspapers, or any other online and offline media. As one of the deliverables you are going to write a short article for the University paper Delft in which you will confirm (green-do) or refute (green-wash) this claim based on facts and figures using argumentation.	
Study Goals	After following this course, you should be able to: - acquire an overview of recent developments in sustainability. - assess recent developments in a critical and engineering way by quantifying sustainable issues and use scientific literature to underpin or refute claims made about it. - convey your findings concisely using text and visualizations.	
Education Method	We think that learning is best done interacting and learning from and with each other. We have developed an approach where we meet each other regularly during the Discussion sessions to discuss the progress of your claim investigation. We have setup several lectures, online content and workshops where we will introduce you to fact checking, substantiating findings, data visualization, argumentation, and storytelling. For this we invited several experts from different faculties at the TU Delft.	
Assessment	At the end of the course, you have two deliverables: a written piece of 500 to 600 words like in the form of NRCcheckt, plus appendices with the evidence. Your article plus its appendices will be reviewed on the following topics: - Clarity of the claim, background, and reference - Research questions & approach - The use and reliability of literature to substantiate your findings - The use and reliability of calculations to substantiate your findings - The use and reliability of tests and measurements to substantiate your findings - Use, quality, and the clarity of figures to substantiate your findings - Structure, brevity, clarity, and convincingness of the story - Soundness of the conclusions and use of argumentation and reasoning refuting or supporting the claim - Appendices showing hand calculations, references, etc.	

Year	2022/2023
Organization	Industrial Design Engineering
Education	Minors Industrial Design Engineering

IO-MI-223

Program Title	Connected Creativity
Program Coordinator	Milene Gonçalves (m.guerreirogoncalves@tudelft.nl)
EC Program	30
Program Start (Study Year)	2020
Program Goals	The ultimate goal of Connected Creativity is to support the development of future-ready innovators to approach challenges with deliberate creativity in an interdisciplinary context. Connected Creativity is split up in two parts: Q1 Ignite and Q2 Electrified. By the end of Q1 Ignite, students will be able to: - Evaluate best practices in relation to individual and teams creative path - Research and compare different creativity theories and strategies. - Apply creativity attitudes and skills, working both in teams and individually. - Apply a deliberate creative process, methods and techniques, in all stages of a creative problem solving process in an interdisciplinary context. By the end of Q2 Electrified, students will be able to: - Apply a deliberate creative process in a real, interdisciplinary context, by connecting creativity and innovation in organisations. - Indicate success factors and barriers of creativity in organizations. - Investigate how creativity is situated in social, political and historical contexts. - Criticise different scientific perspectives on creativity.
Intended for	This minor is for those who are interested in applying deliberate creativity in an interdisciplinary context. The course asks for a certain level of curiosity which we expect all students to have, but prior knowledge on creative processes is not necessary. As we aim for an interdisciplinary approach, we hope to welcome students from different disciplines and universities.
Minor Content 1	In Q1 Ignite, the students will learn the basics of creativity through three courses. Half of the time (7.5 EC) they will spend on Deliberate Creativity (DC). This is a practical course in which the students learn about the iCPS-process (Integrated Creative Problem Solving) through real cases from different domains and topics. The focus of this course is on creativity skills and attitudes. CreativiTree (CT) is a more knowledge-based course in which the students will learn theoretical backgrounds of creativity strategies. This course accounts for 4.5 EC. They can apply this knowledge in the aforementioned course DC. The Creative Path (CP) is a course with the focus on reflection and the development of skills from the individual to the team level. This course will be 3 EC. Students will learn theoretical models on reflection, by practicing reflection moments at different levels (individual, team, organization, culture) throughout Ignite. This course supports the other two courses in capturing knowledge, skills and attitudes and will help the students in becoming creative individuals. The CP course starts with an introduction week: a Creativity Deepdive. This week will set the stage for the entire Minor.
Minor content 2	Q2 Electrified is divided in two courses: the Practice of Creativity (PoC) and Contextualising Creativity (CC). In PoC all the knowledge from Block A will be deployed in an organizational context. The link between creativity and innovation will be made. Special focus will be on barriers, success factors and the urgency of creativity within organizations. Real organizations will be involved in the course to share real life cases. And finally, Contextualising Creativity (CC) situates the emergence of creativity theories/practices in social, political and technological contexts. Students investigate how creativity-theories came about in the context of industrialization, expectations of artificial intelligence, and the Cold War. CC is driven by the students curiosity and ambition to learn more about the creative field.
Minor Assessment	The grading of the assignments will be conducted by lecturers and coaches of the course. Benchmarking will take place in the case of multiple, parallel coaches and between courses to ensure consistency. All assignments will be graded using a rubric. Q2 cannot be started without active participation in Q1.
Maximum number of participants	40
Remarks Maximum number of participants	All disciplines are welcome in Connected Creativity. From engineering to psychology. From German literature to astrophysics. We aim for a division between students based on: 1/3 from the faculty of Industrial Design Engineering (TU Delft) 1/3 from Engineering disciplines (TU Delft) 1/3 other disciplines 40 students is the maximum we accept. If there are more applications we will use a random draw to place students. Minors of the IDE faculty are open only for students from academic programs.

Dr.ir. S.F.J. Flipsen

Unit	Industrieel Ontwerpen
Department	Circular Product Design
Telephone	+31 15 27 89398
Room	32.B-3-060

ontbreekt

Ir. C.P.J.M. Kroon